

Natural Language Processing is Fun!

How computers understand Human Language



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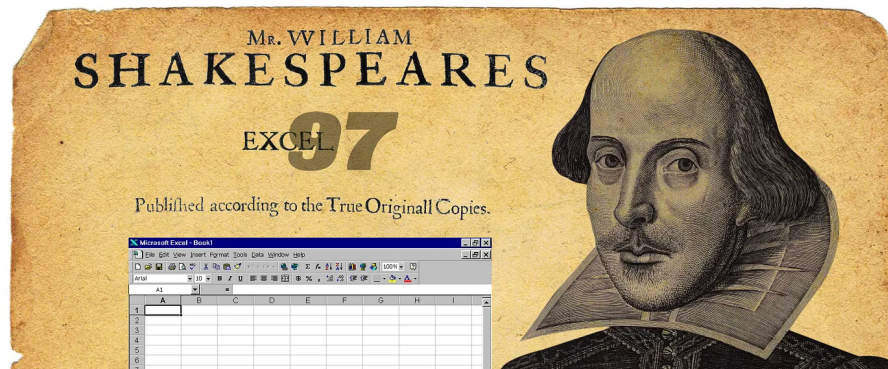
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This article is part of an on-going series on NLP: [Part 1](#), [Part 2](#), [Part 3](#), [Part 4](#), [Part 5](#). You can also read a reader-translated version of this article in [普通话](#) or [فارسی](#).

Giant update: *I've written a new book based on these articles! It not only expands and updates all my articles, but it has tons of brand new content and lots of hands-on coding projects. [Check it out now!](#)*

Computers are great at working with structured data like spreadsheets and database tables. But us humans usually communicate in words, not in tables. That's unfortunate for computers.



Unfortunately we don't live in this alternate version of history where all data is structured.

A lot of information in the world is unstructured — raw text in English or another human language. How can we get a computer to understand unstructured text and extract data from it?

London is the capital and most populous city of England and the United Kingdom. Standing on the River Thames in the south east of the island of Great Britain, London has been a major settlement for two millennia. It was founded by the Romans, who named it Londinium. London's ancient core, the City of London, largely retains its 1.12-square-mile (2.9 km²) medieval boundaries.

Q 1

Natural Language Processing, or NLP, is the sub-field of AI that is focused on enabling computers to understand and process human languages. Let's check out how NLP works and learn how to write programs that can extract information out of raw text using Python!

Note: If you don't care how NLP works and just want to cut and paste some code, skip way down to the section called "Coding the NLP Pipeline in Python".

Can Computers Understand Language?

As long as computers have been around, programmers have been trying to write programs that understand languages like English. The reason is pretty obvious — humans have been writing things down for thousands of years and it would be really helpful if a computer could read and understand all that data.

Computers can't yet truly understand English in the way that humans do — but they can already do a lot! In certain limited areas, what you can do with NLP already seems like magic. You might be able to save a lot of time by applying NLP techniques to your own projects.

And even better, the latest advances in NLP are easily accessible through open source Python libraries like [spaCy](#), [textacy](#), and [neuralcoref](#). What you can do with just a few lines of python is amazing.

Extracting Meaning from Text is Hard

The process of reading and understanding English is very complex — and that's not even considering that English doesn't follow logical and consistent rules. For example, what does this news headline mean?

| *“Environmental regulators grill business owner over illegal coal fires.”*

Are the regulators questioning a business owner about burning coal illegally? Or are the regulators literally cooking the business owner? As you can see, parsing English with a computer is going to be complicated.

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